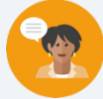


12.1 Creating Proportions

Lesson Introduction

	Benchmark	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.			Learning Target	I can create proportions to represent a situation.
	Benchmark Coherence	Building On MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.		Working Toward MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.		
	Essential Questions	- How do you use proportions to represent a situation and find missing information? - What role does unit rate play in proportional relationships?				
	Lesson Narrative	This lesson formally introduces creating proportions to represent situations, focusing on how multiple proportions can represent a single situation and that solving proportions reveals different information about the initial problem. This lesson deeply dives into a single-problem context to explore the different ways proportions can be set up, moving slowly from procedures to concepts before releasing students to a different context in the 12.1.6 Collaboration. The contexts used in this lesson spiral content from previous units to expose students to additional proportional reasoning about those contexts.				
	Component Summary	12.1.1 Warm-Up (~5 min.)	Consider a Math Flop (<i>SE p. 213</i>)	12.1.4 Guided Instruction (5 min.)	Study of the structure of proportions using length units (<i>SE p. 215</i>)	
		12.1.2 Exploration (5-10 min.)	Consider solution strategies in estimating with scale factors (<i>SE p. 214</i>)	12.1.5 Guided Practice (5-10 min.)	Encounter contextualized proportion building with whole numbers (<i>SE p. 216</i>)	
		12.1.3 Bring It Together (5 min.)	Define “proportional relationship,” “proportion,” and “unit rate” (<i>SE p. 215</i>)	12.1.6 Collaboration (10-15 min.)	Apply methods of solving with proportions (<i>SE p. 216</i>)	
		12.1.7 Wrap-Up (~5 min.)	Analyze the proportions of three students to find the errors (<i>SE p. 217</i>)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - Proportional relationship - Proportion - Unit rate <u>Additional Terms:</u> N/A		N/A		N/A

 Teacher Tips	Common Misconceptions • Students may struggle to interpret proportions that are structured in a way that is different from their original thinking. • Students may focus on one quantity instead of the relationship between quantities. • Students may confuse the glossary terms of rate, ratio, and proportion. • Students may try to solve proportions using additive reasoning rather than multiplicative thinking.	Things to Consider • Instruction should avoid referencing a “correct” way to set up proportions and instead highlight multiple pathways. • Students may attempt to use other incorrect proportions as a check for whether another incorrect proportion is viable or not. Both will result in an incorrect value, so the focus for this lesson should be on ensuring the ratios are correct before calculating. • Students should be encouraged to label the ratios they use in their proportions to ensure they are setting them up in a way that will result in accurate, equivalent ratios.
	Literacy and Language Routines Think-Pair-Share Utilize a Think-Pair-Share for 12.1.2 Exploration questions 2 and 3. Give students time to make observations and find the similarities and differences independently before sharing their observations with a partner.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students will be working to represent the relationships between values within a problem context. Starting in 12.1.2 Exploration, highlight the process by which students are moving from understanding connections between their comprehension of the problem and representing that as a unit rate versus with equivalent ratios in 12.1.4 Guided Instruction and then the connection to creating a proportion in 12.1.6 Collaboration.
	 Differentiate Instruction	
Lesson Notes:		

12.1.1 Warm-Up: Math Flop

Component Overview: Students make observations on math in a real-world context, specifically targeting the error in algebraic and visual representations.



12.1.1 Warm-Up: Math Flop

The advertisement encourages making DiGiorno pizza a part of your weekly meals.



For students who might not notice the Math Flop:

- How much is one portion?
- What is the visual representation of the pizza trying to communicate?
- Which is more: $\frac{1}{5}$ or $\frac{1}{4}$?

1. What do you notice about the advertisement?

Answers will vary. It says one portion of the pizza is $\frac{1}{5}$ but the image shows fourths.

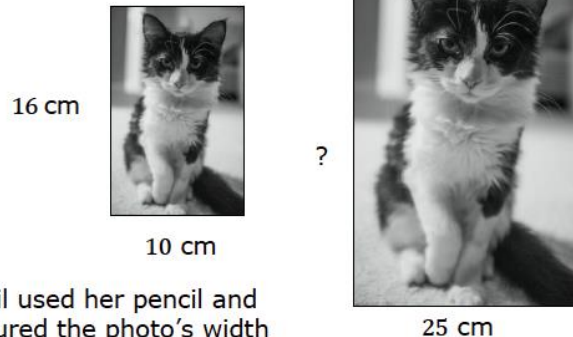
12.1.2 Exploration: Proportional Thinking

Component Overview: Students consider the measuring tools and processes a student uses to estimate a missing height of an enlarged poster from a photograph. Then students analyze two different ways students used to calculate the missing height measurement using proportional reasoning. From these examples, students describe the similarities and differences between the calculation strategies.



12.1.2 Exploration: Proportional Thinking

A photograph is enlarged to make a poster. Measurements are given in centimeters (cm).



Abigail used her pencil and measured the photo's width using her finger to mark off 10 cm on the pencil. She then used the pencil to estimate the height using the mark on her pencil.

1. Discuss with your partner why Abigail's strategy could lead to an incorrect answer.

Answers may vary. Abigail's strategy may be useful for estimating, but not for accurate or precise measurements. Abigail did not use a ruler or equivalent tool to measure the photograph.



The Exploration purposefully creates the need for a proportion before trying the calculations. Monitor for student responses that reflect proportional reasoning to share with the whole group.

For example, students might say, "Abigail could do that, but her answer might be a little bit off because her measurement will not be as precise."

Would Abigail's strategy work for finding the distance between two cities on a map? Why or why not?

12.1.2 Exploration: Proportional Thinking (continued)

Mariana and Yousef used different strategies to find the height of the poster. Review their solution strategies. Then, work with your partner to answer the questions that follow.

Mariana's Work	Yousef's Work
$\frac{25 \text{ cm}}{10 \text{ cm}} \div \frac{10}{10} = \frac{2.5}{1} = 2.5$	$\frac{\text{height}}{\text{width}} \quad \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}}$
Scale factor = 2.5 $16 \text{ cm} \times 2.5 = 40 \text{ cm}$	$\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$
The height of the poster is 40 cm.	The height of the poster is 40 cm.

2. How are Mariana and Yousef's solution strategies similar?

*Actual student answers will vary.
Both multiplied 16 by 2.5 to find 40 cm.
Both got the same answer.*

3. How are their solution strategies different?

*Actual student answers will vary.
Mariana found a unit rate (scale factor).
Yousef set up a proportion.*



Think-Pair-Share

Sample routine strategy:

- *Think:* Take two minutes to make observations about Mariana's work and Yousef's work. What is the same? What is different? Make at least two observations for what is the same and at least two observations for what is different. There are no wrong answers—the more creative, the better!
- *Pair:* Turn to your shoulder partner and share your similarities and differences. Place a check mark next to any that are the same and circle the ones that are different.
- *Share:* Share one similarity or difference that your partner had that was unique.

Monitor for student responses in the Pair that highlight learning targets.

12.1.3 Bring It Together: Proportional Relationships

Component Overview: Students are introduced to the definitions for proportional relationships, proportions, and unit rates. Students then refer to these definitions and the examples from the previous activity to complete a written description in paragraph form of the two students' work.



12.1.3 Bring it Together: Proportional Relationships



In a **proportional relationship**, the values for one quantity are each multiplied by the same number to get the values for the other quantity.

A **proportion** sets two or more equivalent ratios in such relationships as equal to each other and can be used to solve problems.



A **unit rate** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

1. Use the words and phrases in the bank below to complete the description of Mariana and Yousef's work.

is equal to proportion ratio unit rate

The photograph and the poster are in a **proportional relationship** because the poster is a scaled version of the photograph. Mariana found the scale factor, which is a unit rate, and multiplied it by the height of the photo to find the height of the poster. Yousef set up a proportion stating the ratio of the height to width of the photograph is equal to the ratio of the height to width of the poster.



Consider having students notate and make connections between these terms and Mariana's and Yousef's work in the previous activity, as shown in the key.



How is the concept of proportions similar to or different from what you have done in previous lessons with:

- money?
- weight and mass?
- volume?

Consider referencing Abigail's work in the 12.1.2 Exploration. What would you tell Abigail now?

12.1.4 Guided Instruction: Proportional Thinking

Component Overview: Within the same problem context, students encounter four different strategies to create a proportion that can be used to find the missing enlarged height. Students then identify from these completed examples the scale factor between the photograph and the poster as well as the scale factor between the height and the width.



12.1.4 Guided Instruction: Proportional Thinking

Proportions can be set up in different ways using the relationships between each quantity.

Four different proportions that can be used to find the height of the poster are shown.

A photo to poster $\frac{\text{height}}{\text{width}} \left \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}} \right.$ $\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$	B poster to photo $\frac{\text{width}}{\text{height}} \left \frac{25 \text{ cm}}{?} = \frac{10 \text{ cm}}{16 \text{ cm}} \right.$ $\frac{25 \text{ cm}}{40 \text{ cm}} = \frac{10 \text{ cm}}{16 \text{ cm}} \times \frac{2.5}{2.5}$
C width to height $\frac{\text{poster}}{\text{photo}} \left \frac{10 \text{ cm}}{25 \text{ cm}} = \frac{16 \text{ cm}}{?} \right.$ $\frac{10 \text{ cm}}{25 \text{ cm}} \times \frac{1.6}{1.6} = \frac{16 \text{ cm}}{40 \text{ cm}}$	D height to width $\frac{\text{photo}}{\text{poster}} \left \frac{?}{16 \text{ cm}} = \frac{25 \text{ cm}}{10 \text{ cm}} \right.$ $\frac{40 \text{ cm}}{16 \text{ cm}} = \frac{25 \text{ cm}}{10 \text{ cm}} \times \frac{1.6}{1.6}$

All of the proportions show that the ratios of the dimensions of the photograph and the dimensions of the poster are equivalent.

Complete the statements based on the proportions.

- The dimensions of the poster are

☐ 1.6
☒ 2.5

 times larger than the photograph.
- The height of both the photograph and poster is

☒ 1.6
☐ 2.5

 times larger than the width.



Number Talk

Consider a Number Talk routine to have students notice the similarities and differences between each proportion, ultimately concluding that all proportions can be used (flexible pathways).

Compare and contrast each method. Make note of three things that are the same and one thing that is different or unique to each method.



Consider Four Corners for a kinesthetic activity, with students discussing at each corner which proportion makes the most sense to them.

12.1.5 Guided Practice: Proportional Thinking

Component Overview: In this component, students are provided with a new context, moving from a sample to a population using proportions. Starting with identifying the proportional relationship between the households (with dogs and total) and sample to population. Students then compare two proportions, correcting the incorrect one and choosing one to use to solve.



12.1.5 Guided Practice: Proportional Thinking

The city of Tampa collected sample data on pet ownership to use when determining whether or not it needed to build additional dog parks. A city parks officer randomly sampled 200 households in Tampa and found that 126 of the households had a dog. Using the results of the sample, determine how many households have a dog if there are 153,918 households in Tampa.

- Complete the statements to describe the relationships that are being compared.
 - The number of households with a dog is being compared to the total number of households.
 - The data sample is being compared to the population of Tampa.
- Two proportions that represent this situation are below, but one of them has an error.

Proportion A			Proportion B		
<i>Sample</i> <i>Dogs</i>	126	<i>Pop</i> 153,918	<i>Total</i> <i>Population</i>	153,918	<i>Dogs</i> <i>Sample</i>
<i>Total</i>	200	$\frac{153,918}{x}$		$\frac{153,918}{200}$	$\frac{x}{126}$

Proportion ☐ A ☐ B is not set up correctly. The proportion should be $\frac{126}{200} = \frac{x}{153,918}$.

- Find the number of households expected to have a dog.
≈ 96,968 households
Teachers can ask students why it makes sense to round the households to the nearest whole number. Potential answer: Households involve people and are not measured in fractions or parts.



Notice the embedded scaffolding in questions 1-3.



How else could this proportion have been correctly set up? (Using similar thinking to the previous component.)



For question 3, have students discuss why we needed to round the number of households.

12.1.6 Collaboration: Creating Proportions

Component Overview: Students continue to write two possible proportions given the problem context and pick one to use in the context of speed and a scale drawing.



12.1.6 Collaboration: Creating Proportions

1. Javad drove his car to Canada. He was driving on a road and saw the sign shown to the right. According to his speedometer, Javad was driving 75 miles per hour when he was pulled over for speeding. Canadian speed limits are given in kilometers (km) per hour because they use the metric system. There are approximately 1.61 kilometers in 1 mile.



Notice the contexts used in this lesson are familiar contexts from previous lessons or units, including scale drawings and using data to draw conclusions about populations.

- a. Create two different proportions that could be used to find Javad's speed in kilometers per hour.
Answers may vary.

Proportion A	Proportion B
$\frac{1.61 \text{ km}}{1 \text{ mi}} = \frac{x \text{ kmph}}{75 \text{ mph}}$	$\frac{x}{1.61 \text{ km}} = \frac{75 \text{ mph}}{1 \text{ mi}}$

- b. How fast was Javad traveling in kilometers per hour?

$$\frac{75 \cancel{\text{mi}}}{1 \text{ hr}} \times \frac{1.61 \text{ km}}{1 \cancel{\text{mi}}} =$$

$$120.75 \text{ km/h}$$

12.1.6 Collaboration: Creating Proportions (continued)

2. An architect made a scale drawing of a house as shown.



The scale for the drawing is $\frac{1}{4}$ inch = 6 feet.

- a. Create two different proportions that could be used to find the actual length of the house represented by the drawing.

Answers will vary.

Proportion A	Proportion B
$\frac{\frac{1}{4} \text{ in.}}{6 \text{ ft.}} = \frac{3.5 \text{ in.}}{x}$	$\frac{3.5 \text{ in.}}{\frac{1}{4} \text{ in.}} = \frac{x}{6 \text{ ft.}}$

- b. What is the length of the house represented by the drawing?

84 ft.



Consider prompting students while monitoring to note rounding precision:

- How does the context inform how you are rounding?
- Would this be the same if there was no context, only calculations? Why or why not?

12.1.7 Wrap-Up: Error Analysis

Component Overview: This formative assessment helps you identify whether students can relate the context of the problem to the structure of the proportion. Use this data to inform which questions and supports you should emphasize in the next lesson.



12.1.7 Wrap-Up: Error Analysis

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. = 90 miles (mi).

Three students set up proportions to find the actual distance between Jacksonville and Gainesville, but two of them made errors.

- For each proportion, check whether it is correct or incorrect. If it is incorrect, fix the error by changing one of the ratios in the proportion.

Lily's Proportion	Eli's Proportion	Mia's Proportion
$\frac{2}{1.4} = \frac{90}{1.4x}$	$\frac{1.4}{2} = \frac{90}{x}$	$\frac{1.4}{x} = \frac{2}{90}$
☐ correct ☒ incorrect	☐ correct ☒ incorrect	☒ correct ☐ incorrect



For homework, consider having students to write an explanation of what they changed and why as if they were explaining it to a friend using the vocabulary words from 12.1.3 Bring It Together.